

6(a)(i)	<u>waves</u> at (each) slit/aperture spread	B1
	(into the geometric shadow) <u>wave(s)</u> overlap/superpose/sum/meet/intersect	B1
6(a)(ii)	there is not a constant phase difference/coherence (for two separate light source(s)) or waves/light from the double slit are coherent/have a constant phase difference	B1
6(b)	$x = \lambda D / a$	C1
	$\lambda = (36 \times 10^{-3} \times 0.48 \times 10^{-3}) / (16 \times 2.4)$	C1
	$= 4.5 \times 10^{-7} \text{ m}$	A1
6(c)(i)	<u>no</u> movement of the water/water is flat/no ripples/disturbance	B1
	the path difference is 2.5λ or the phase difference is 900° or 5π rad	B1
6(c)(ii)	1. surface/water/P vibrates/ripples and as (waves from the two dippers) arrive in phase	B1
	2. surface/water/P vibrates/ripples and as amplitudes/displacements are no longer equal/do not cancel	B1